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Number Puzzles and Age Problems

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Number Puzzles and Age Problems

A number puzzle gives clues to help you find an unknown number. To solve number puzzles, you need to understand the words that indicate operations.

sum →	difference →	product →	quotient →
add	subtract	multiply	divide

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Example 1

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Twelve less than the product of a number and 3 is 4 more than twice the number. What is the number? $(=)$

Step 1

Let n equal the unknown number. The clue describes two series of operations that can be performed using n . The word *is* tells us that the results of each series are equal.

Write each part of the sentence as an expression.

- 12 less than the product of a number and 3

$$3n - 12$$

- 4 more than twice the number

$$2n + 4$$

Step 2

Write and solve an equation.

$$3n - 12 = 2n + 4$$

Add 12 to both sides.

$$3n - 12 + 12 = 2n + 4 + 12$$

$$3n = 2n + 16$$

Subtract $2n$ from both sides.

$$3n - 2n = 2n - 2n + 16$$

$$n = 16$$

The number is **16**.

After finding an answer, good test takers read the problem one more time to make sure the answer makes sense.

Category

Lesson 6: Common Algebra Word Problems

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"10% of" means $(0.10) \times$ —



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Example 2[Hide](#)

Fred is 5 times as old as his grandson Joe. In 10 years, the sum of their ages will be 92. How old is Fred now?

Step 1

Make a chart to organize the facts.

- Let x represent Joe's age now.
- Fred's age now is 5 times Joe's.
- In the future, they will each be 10 years older than they are now.

	Age Now	Age in 10 Years
Joe	x	$x + 10$
Fred	$5x$	$5x + 10$

Step 2

Write and solve an equation. The sum of their ages in 10 years will be 92.

$$(x + 10) + (5x + 10) = 92$$

$$6x + 20 = 92$$

$$6x + 20 - 20 = 92 - 20$$

$$6x = 72$$

$$x = 12$$

Step 3

Answer the question.

The variable x equals 12. The question asks for Fred's age now, which is $5x$ Joe's age. Fred is $5(12) = 60$

Fred is 60 years old.

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